

RESEARCH PAPER

Adoption and maintenance of a Mediterranean diet in patients with coronary heart disease from a Northern European population: a pilot randomised trial of different methods of delivering Mediterranean diet advice

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Abstract

Background: A Mediterranean diet has been shown to protect against coronary heart disease (CHD). Adherence to a Mediterranean diet can be assessed using a Mediterranean diet score. The primary aim of this pilot study was to examine whether CHD patients in a Northern European population would adopt and maintain a Mediterranean diet, with a secondary aim of comparing the effectiveness of different methodologies aimed at improving compliance.

Methods: Sixty-one patients with a diagnosis of CHD were randomised to one of three groups: either to receive conventional dietetic advice for CHD or advice to implement a Mediterranean-style diet using either behavioural counselling or nutritional counselling. Patients received a follow-up assessment at 6 months (adoption) and a subset of patients was followed up at 12 months (maintenance). The primary outcome measure was the between-group difference in the mean change in Mediterranean diet score (MDS).

Results: The change in MDS was not significantly different between groups. However, all three groups reported a significant within-group increase in MDS ($P < 0.01$) at 6 and 12 months follow-up.

Conclusions: All three groups made dietary changes towards a Mediterranean diet, but behavioural counselling did not have significant additional benefit over nutritional counselling in initiating and maintaining dietary change, and neither method offering specific Mediterranean diet advice had any significant benefit in terms of improvement in MDS over conventional dietetic practice.

Introduction

Cardiovascular disease (CVD) remains a leading cause of morbidity and mortality in the Western world. Adherence to a Mediterranean diet (MD) has been associated with a reduced risk of CVD (Panagiotakos *et al.*, 2008), diabetes (Martinez-Gonzalez *et al.*, 2008) and overall mortality in both European (Trichopoulos *et al.*, 2003, 2005; Knoop *et al.*, 2004) and non-European (Mitrou *et al.*, 2007) populations, and this has recently been subjected to meta-analysis (Sofi *et al.*, 2008), showing that greater adherence to a MD was associated with a significant improvement

in health status, whether assessed as overall mortality, CVD mortality, incidence of or mortality from cancer or incidence of Parkinson's disease and Alzheimer's disease. Similarly, a systematic review of the evidence supporting a causal link between dietary factors and CVD rated the evidence for the MD as being strong (Mente *et al.*, 2009).

A number of short-term interventions have examined the effect of the MD on cardiovascular risk factors (Vincent-Baudry *et al.*, 2005; Estruch *et al.* 2006) and metabolic syndrome status (Esposito *et al.*, 2004; Salas-Salvado *et al.*, 2008), but, to date, only one study has reported clinical endpoints. The Lyon Diet Heart Study

examined the effect of a MD in survivors of a first myocardial infarction (MI), observing a significant reduction in major coronary events (de Lorgeril *et al.*, 1994, 1996, 1999). The results of the Lyon study (de Lorgeril *et al.*, 1999) were obtained in a Mediterranean population, and limited evidence is available about whether such a diet would be acceptable or effective in Northern European populations, whose baseline dietary habits are likely to be very different to those of the Lyon population.

Adherence to the MD can be assessed using a Mediterranean diet score (MDS), which summarises overall diet using a single score resulting from consideration of a number of different dietary components (including foods, food groups or a combination of foods and nutrients). A number of these have been developed, and these have been comprehensively reviewed (Bach *et al.*, 2006).

The primary aim of this pilot study was to determine whether coronary heart disease (CHD) patients in a Northern European population would adopt and maintain such a MD, with a secondary aim of comparing the effectiveness of different methodologies aimed at improving compliance. Two different approaches to dietary change were compared: standard nutritional counselling and behavioural counselling. Behavioural counselling has recently been shown to be effective in producing sustained increases in fruit and vegetable consumption in low-income adults in the general population (Steptoe *et al.*, 2003), and for reducing fat intake and cigarettes smoked and increasing regular exercise in patients at increased risk of CHD (Steptoe *et al.*, 1999).

Materials and methods

Patients and study design

This study was a pilot randomised controlled parallel group trial. A total of 61 participants were recruited from the Cardiology Directorate, Royal Victoria Hospital, Belfast. Patients recruited to the study had a recent diagnosis (within past 4 weeks) of MI or unstable angina. Recruitment took place over a 12-month period, from December 2004 to December 2005. Those patients with severe heart failure, awaiting urgent inpatient coronary revascularisation, on warfarin therapy, on OmacorTM therapy, with a history of excessive alcohol ingestion, with a life expectancy of less than 6 months for other reasons, with cognitive impairment, with an inability to give informed consent or comply with the diet or already taking multivitamin/fish oil supplements, were excluded from the study.

Ethical approval was obtained for this study from the Research Ethics Committee of Queen's University Belfast. Informed written consent was obtained from all participants. Willing participants ($n = 61$) were randomised

(using a block randomisation approach with computer generated random numbers) to one of three arms: either conventional dietetic advice (CDA) for CHD or advice to implement a MD using nutritional counselling (MDNC) or advice to implement a MD using behavioural counselling (MDBC). All dietary advice in all treatment groups was given by the same research dietitian, and quantitative recommendations were made wherever possible in terms of frequency of servings.

Dietary interventions

CDA group

Those patients in the control arm received conventional dietary advice during admission and at the Cardiac Rehabilitation classes as per normal current practice in the unit, and as summarised in the form of a diet sheet given to the participant by the research dietitian at recruitment. They then received no further contact from the research dietitian until the month 6 and 12 follow-up appointments. This advice was in the form of general advice to adopt a low-fat, cardioprotective diet.

MDNC group

This experimental group received detailed information regarding the implementation of the MD and participants were provided with a diet sheet, which contained detailed advice and information on the MD, the potential health benefits of the diet, recipe ideas and a sample meal plan. These participants then received a home visit from the research dietitian at week 1 (optional), and further home visits at months 1, 2 and 4. Patients were also able to contact the researcher for advice by telephone during normal working hours.

MDBC group

This experimental group had a similar number of contacts with, and visits from, the research dietitian as the MDNC group but, this time, behavioural counselling was used to deliver the dietary advice. The behavioural counselling method is founded on social learning theory and the stage of change model, which posits that the most appropriate methods of encouraging change in behaviour vary with the motivational readiness of the individual (Prochaska & Velicer, 1997). Interventions were therefore tailored to the individual, with personal specific advice and setting of short- and long-term goals based on their stage of change measure, which reflected their readiness to adopt a MD. Participants' stage of change was assessed using a modified version of a previously validated questionnaire, which was used in a study that aimed to achieve increases in fruit and vegetable consumption (Steptoe *et al.*, 2003). Patients in the behavioural

counselling group were provided with the same diet sheet as those in the nutritional counselling group but, in addition, were also provided with a 'Help to Change' booklet. This contained a list of the common difficulties found when making dietary change, as well as suggestions for overcoming these.

Details of Mediterranean interventions (with nutritional or behavioural counselling): Patients in each MD group were advised to adopt a diet similar to that described by de Lorgeril *et al.* (1994). Patients were advised to consume seven to 10 portions of fruit and vegetables each day (including more root vegetables and green vegetables), more wholegrain cereals, more fish (four portions per week), less meat (approximately once per week) (beef, lamb and pork were to be replaced by poultry), and butter and cream were to be replaced with an olive-oil based spread. The oils recommended for salads and food preparation were olive and rapeseed oils exclusively. Moderate alcohol consumption, in the form of wine, was allowed at meals. Patients were also advised to include unsalted nuts as snacks.

The key differences in the advice given to the CDA group compared to the MD groups were that the MD groups were given specific advice to use olive and rapeseed oils exclusively, whereas those in the CDA group were simply advised to replace saturated fats with mono- or poly-unsaturated fat. In addition, those in the MD groups were advised to include more fruit and vegetables, fish and nuts and less red meat than those in the CDA group. The advice given to patients in the CDA group was similar to that provided to those in the MD groups in terms of oily fish intake (increase to two to three portions per week) and intake of wholegrain cereals.

Data collection/endpoint assessment

Baseline demographic information including smoking status was recorded. Weight, height and blood pressure were assessed, a MDS (Martinez-Gonzalez *et al.*, 2004) calculated and a fasting blood sample collected, separated and stored at -80°C . Patients were then asked to attend the hospital for a further, similar assessment at month 6, whereas a subset of the study population ($n = 36$) were also followed up at month 12

MDS

A MDS was calculated using a validated questionnaire, as described by Martinez-Gonzalez *et al.* (2004). This questionnaire assessed the frequency of consumption, for a typical serving, of nine food items in the MD. One point was added for a high intake of each food that positively contributed to a MD and for a low intake of foods that

negatively contributed to a MD. The resulting score was in the range 0–9 points, with higher scores indicating greater compliance with the MD.

Laboratory analysis

Plasma vitamin C was assessed by fluorimetric assay using a Cobas FARA analyser (Roche Diagnostics, Basel, Switzerland) (Vuilleumier & Keck, 1989). Plasma fatty acids were assessed by gas chromatography (Morrison & Smith, 1964).

Study endpoint

The primary endpoint of the study was the between-group difference in the mean change in MDS after 6 months follow-up.

Statistical analysis

Statistical analysis was carried out using SPSS for Windows, version 13 (SPSS Inc., Chicago, IL, USA). Variables were tested for normality and were logarithmically transformed where necessary. One-way analysis of variance (ANOVA) was used to compare the three intervention groups at baseline. The difference in change in MDS between groups was compared using one-way ANOVA with suitable post-hoc comparison tests. The within-group change in the parameters measured (pre- versus post-intervention) was compared using paired samples *t*-tests.

Because this was a pilot study aiming to examine whether patients would comply with dietary advice and there were no relevant data available for the population in Northern Ireland, power/sample size calculations were not considered to be appropriate.

Results

Participant flow and characteristics

The participants recruited to the study were aged between 39–78 years. A total of 61 patients were randomised; however, three were excluded after randomisation because of changes in treatment plans, meaning they were no longer eligible to continue in the study. Therefore, baseline data are available for 58 participants. Forty-nine participants were followed up for 6 months because nine participants (15.5%) withdrew from the study before their 6-month follow-up (the reason for withdrawal was not recorded). Figure 1 summarises the flow of participants through the study.

The mean age of the study population was 57.1 years, and the majority of participants were male (81%). Mean blood pressure at baseline was normal (124/68 mmHg),

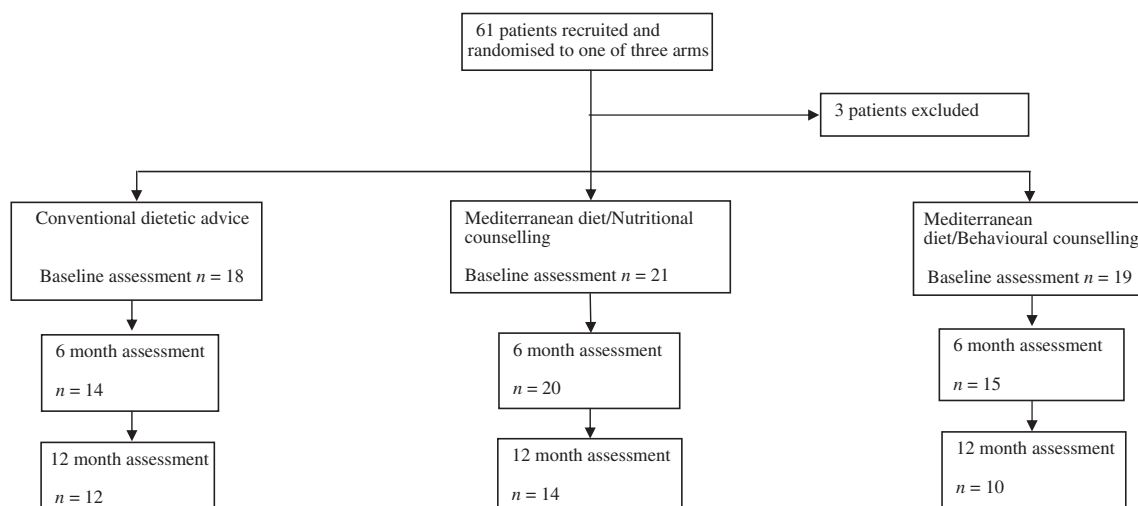


Figure 1 Flow of participants through the study protocol.

although many of these patients were already on anti-hypertensive treatment at baseline. Mean body mass index was 30.1 kg m^{-2} , with 84.5% classified as being overweight and 44.8% classified as obese. At baseline, 20.7% of participants were current smokers.

Comparison of baseline data by group

The baseline characteristics of the three intervention groups are shown in Table 1.

Stage of change for a MD was assessed in the MDNC group at baseline ($n = 19$). Thirteen (68%) participants were found to be in preparation, whereas three (16%) were in contemplation and three (16%) were in maintenance.

Change in MDS

Baseline data on MDS are presented in Table 1. Data on the change in the score in the three groups at 6- and 12-month follow-up is presented in Table 2. There was no significant difference between the groups for change in MDS at 6 months ($P = 0.365$) and 12 months ($P = 0.860$). However, there was a significant increase in MDS within each of the three groups at 6 months, as well as in the subset of patients followed up at 12 months.

Change in biochemical variables

Baseline data on biochemical variables [vitamin C, oleic acid status and eicosapentaenoic acid (EPA) status] is

Table 1 Comparison of baseline characteristics by intervention group

Characteristic	CDA ($n_{\text{max}} = 18$)		MDNC ($n_{\text{max}} = 21$)		MDBC ($n_{\text{max}} = 19$)	
	Mean	(SD)	Mean	(SD)	Mean	(SD)
Age (years)	55.6	(10.9)	57.7	(7.8)	58.0	(9.2)
Sex: male (%)	77.8	–	81.0	–	84.2	–
BMI (kg m^{-2})	29.2	(5.1)	31.8	(5.7)	29.1	(5.5)
Smoking status: smoker* (%)	22.2	–	19.0	–	21.1	–
Mediterranean diet score	2.33	(1.61)	3.57	(1.66)	3.32	(1.60)
Vitamin C ($\mu\text{mol L}^{-1}$)	27.0	(26.4)	32.9	(21.5)	22.8	(16.1)
Oleic acid [†]	12.5	(1.4)	12.2	(1.2)	12.4	(1.4)
EPA [†]	2.5	(3.2)	1.5	(1.3)	1.7	(2.2)

BMI, body mass index; CDA, conventional dietetic advice group; MDNC, Mediterranean diet/nutritional counselling group; MDBC, Mediterranean diet/behavioural counselling group; EPA, eicosapentaenoic acid.

Data are presented as mean (SD), except sex and smoking status, which are presented as a percentage.

*Participants were defined as nonsmokers if they had stopped smoking prior to admission to hospital.

[†]Data presented as percentage of total phospholipid fraction

Continuous variables were compared between intervention groups using one-way analysis of variance, and categorical variables by chi-square.

There were no significant differences in baseline characteristics between intervention groups.

Table 2 Change in Mediterranean diet score at 6 months and 12 months in the three intervention groups†

Time	CDA		MDNC		MDBC		P-value‡
	Mean	CI	Mean	CI	Mean	CI	
6 months	2.07** (n = 14)	1.05, 3.10	2.70** (n = 20)	1.95, 3.46	2.87** (n = 15)	2.15, 3.59	0.365
12 months	2.67** (n = 12)	1.25, 4.08	2.43** (n = 14)	1.50, 3.35	2.80** (n = 10)	2.24, 3.36	0.860

CDA, conventional dietetic advice group; MDNC, mediterranean diet/nutritional counselling group; MDBC, mediterranean diet/behavioural counselling group.

Data are presented as the mean change and 95% confidence interval.

†Mean change calculated as 6 or 12 month visit – baseline visit.

‡P-values for differences in change from baseline to 6 months or 12 months between the three groups (one-way analysis of variance).

Significant within-group changes (**P ≤ 0.01) from baseline (paired t-test) are shown.

Table 3 Change† in biochemical variables at 6 months and 12 months in the three intervention groups

	Time	CDA		MDNC		MDBC		P-value‡
		Mean	CI	Mean	CI	Mean	CI	
Vitamin C (μmol L ⁻¹)	6 months	6.7 (n = 13)	-8.2, 21.7	14.0** (n = 19)	7.1, 20.9	21.9** (n = 13)	13.5, 30.3	0.09
	12 months	7.9 (n = 11)	-16.0, 31.8	9.1 (n = 14)	-5.5, 23.8	12.5* (n = 10)	2.8, 22.2	0.91
Oleic acid (% of total phospholipid fraction)	6 months	0.01 (n = 12)	-0.49, 0.51	0.81* (n = 15)	0.03, 1.60	0.59* (n = 12)	0.04, 1.14	0.17
	12 months	-0.09 (n = 11)	-0.99, 0.81	0.03 (n = 12)	-0.72, 0.77	0.64 (n = 9)	-0.11, 1.40	0.35
EPA (% of total phospholipid fraction)	6 months	1.12 (n = 12)	-0.69, 2.92	0.83 (n = 15)	-0.21, 1.86	0.53** (n = 12)	0.22, 0.85	0.77
	12 months	0.38 (n = 11)	-0.35, 1.12	0.38 (n = 12)	-0.16, 0.92	0.54 (n = 9)	-0.24, 1.33	0.92

CDA, conventional dietetic advice group; MDNC, mediterranean diet/nutritional counselling group; MDBC, mediterranean diet/behavioural counselling group; EPA, eicosapentaenoic acid.

Data are presented as the mean change and 95% confidence interval.

†Mean change calculated as 6 or 12 month visit – baseline visit.

‡P-values for differences in change from baseline to 6 months or 12 months between the three groups (one-way analysis of variance).

Significant within-group changes (*P < 0.05; **P ≤ 0.01) from baseline (paired t-test) are shown.

shown in Table 1, whereas the changes in biochemical variables in the three groups are presented in Table 3. There were no significant differences between the groups for changes in concentrations of biochemical variables at either 6 months or 12 months, although within-group changes supported the finding of increased MDS, in that vitamin C, oleic acid and EPA all tended to increase.

Discussion

The main aim of the present study was to determine whether Northern European CHD patients would adopt and maintain a MD. The significant within-group change in MDS suggests that all three intervention groups had made changes towards a MD. In the subset of patients followed-up at 12 months, these significant increases were maintained. Although it is difficult to appreciate the significance of this change in MDS, Martinez-Gonzalez *et al.* (2004), who developed the score, have estimated that, for primary prevention, an increment of one point in the score was associated with an 18% reduction in the relative risk of MI. The increases observed in the

current study of 2–3 points are therefore likely to be clinically relevant.

A secondary objective of the present study was to examine whether different methods of delivering dietary advice would affect adoption and maintenance of the MD. There was no significant difference between the groups with regard to the primary endpoint of the study (between-group difference in the mean change in MDS after 6 months follow-up), indicating that behavioural counselling did not appear to have any significant additional benefit over nutritional counselling. Indeed, neither method of dietary intervention involving more specific MD advice and increased contact time with the research dietitian (MDBC and MDNC) had any significant benefit in terms of improvement in MDS over conventional dietetic practice. These pilot data have implications for practice in that usual care appears to be as effective as more intensive dietary interventions, and therefore changes to current practice may not be warranted.

Siero *et al.* (2000) also found that behavioural counselling did not have any significant additional benefit over group nutrition education on Mediterranean nutrition

behaviour. However, behavioural counselling has previously been shown to be effective in achieving increases in fruit and vegetable intake in a study of low-income adults performed by Steptoe *et al.* (2003), as well as in promoting healthy behaviour among adults at increased risk of CHD (Steptoe *et al.*, 1999).

It has previously been reported that those individuals with, or at risk of, disease are more successful at positively changing their diet (Satia *et al.*, 2001; Thomson *et al.*, 2002; Radakovich *et al.*, 2006). The group of patients in the present study had all recently been diagnosed with CHD and perhaps were already motivated to change their diet, regardless of the method used to deliver the dietary advice. In previous studies where behavioural counselling has been used successfully, participants have generally been healthy, and may therefore have required the extra motivation and assistance that behavioural counselling provides to change their behaviour. It is also possible that the patients who consented to participate in this study had an interest in diet and were already motivated to make dietary changes, whereas perhaps those who were less motivated to change were less likely to consent to participate. This possibility is supported by a follow-up to the study by Steptoe *et al.* (2003) on the effect of behavioural counselling to increase fruit and vegetable consumption in low-income adults. Perkins-Porras *et al.* (2005) assessed whether the effect of behavioural counselling varied with stage of change. Behavioural counselling was not effective in promoting further increases in intake among those in the preparation stage, over and above that achieved with nutritional counselling. In the present study, the majority of participants (68%) in the behavioural counselling group were in the preparation stage at baseline. Stage of change was not assessed in the other groups; however, it may be assumed that a similar percentage of participants in the other groups would also have been in the preparation stage at baseline. It is recognised that those in the preparation stage have committed to change, and were therefore motivated to make dietary changes. It is possible that, in the preparation stage, the method of delivering dietary advice was less important than it would be if giving advice to people who were in the earlier stages of change.

Results from the MDS appear to suggest that most of the changes observed at 6 months were maintained in those followed-up at 12 months. Most of the other intervention studies to date which have assessed the MD in relation to biochemical/dietary endpoints have reported a shorter follow-up period and/or maintained dietetic support/contact with the research team for the entire duration of the study with no post-intervention follow-up (Esposito *et al.*, 2004; Vincent-Baudry *et al.*, 2005; Est-ruch *et al.* 2006; Papadaki & Scott, 2008; Salas-Salvado

et al., 2008). Therefore, it is interesting to note that the CHD patients in the present study appeared to maintain dietary changes for at least 6 months when dietetic support was removed. de Lorgeril *et al.* (1999) reported in the final report of the Lyon Diet Heart Study that the protective effect of the MD was maintained up to 4 years after first MI. Again, perhaps the diagnosis of MI/unstable angina gives the necessary motivation not just to make dietary changes, but also to maintain those changes to a greater extent or for a longer period than would be expected in generally healthy participants.

To date, few studies examined what factors influence adherence to a MD, and then only in healthy populations. Sanchez-Villegas *et al.* (2002) showed that being female, older and more physically active were factors associated with greater adherence to a MD, whereas the only factor associated with adherence to a MD pattern in Greek adolescents was actual MD knowledge (Tsartsali *et al.*, 2009). With Greek children and adolescents reporting a more Westernised diet and movement away from the traditional MD (Kontogianni *et al.*, 2008), this could have important public health implications. Although the design of the present study did not allow us to examine these factors and whether they influenced adherence to MD advice, this would be of interest in future appropriately designed studies.

The present study has a number of limitations. This was a pilot study, with small numbers of participants in each arm, and therefore our power to detect differences between intervention groups may have been limited. We compared conventional dietetic practice with specific MD advice, but current conventional dietetic advice is similar in some ways to MD advice (details summarised in the Dietary Interventions section), and therefore the intervention groups may not have been sufficiently different in terms of the actual dietary advice delivered, although they did differ in contact time and how the dietary advice was delivered. The difficulties in designing dietetic intervention studies, particularly in terms of the control group, have recently been reviewed (Baldwin *et al.*, 2008). Finally, the participants who consented to take part in this research study may have already had an interest in diet and have been motivated to make dietary changes, whereas those who were less motivated to change may have been less likely to consent to participate. Unfortunately reasons for nonparticipation were not formally recorded.

In summary, CHD patients in the present study conducted in a Northern European population did make dietary changes towards a MD, but the method of delivering dietetic advice did not affect the degree of change achieved. This may have been because this group of patients were already motivated to make dietary changes

because of their recent diagnosis of MI or unstable angina. This was a pilot study aiming to determine whether CHD patients in Northern Ireland would adopt and maintain a MD. It did not examine whether the dietary advice had any effect on disease outcome measures. Such a study would therefore be required to determine if a MD has additional benefits to pharmacological therapy in the secondary prevention of CHD in a Northern European population.

Conflict of interests, source of funding and authorship

The author declares that they have no conflicts of interest.

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KL was involved in the study design, and carried out all recruitment, dietary analysis, statistical analysis and wrote the first draft of the manuscript. JVW was involved in the study design, advised on statistical analysis and revised the manuscript. ISY was involved in the original study design, advised on interpretation of the data and revised the manuscript. MMcK directed the dietary analysis and revised the manuscript. LPP advised on the use of behavioural counselling. PMcK was involved in the study design, advised on patient recruitment and interpretation of the data, and revised the manuscript. All authors contributed to the concept and design of the manuscript, and critically reviewed and approved the final version submitted for publication.

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